NCEA Level 2 Mathematics

91261 · Apply algebraic methods in solving problems (Algebra)

Excellence-level practice questions

PRACTICE PAST PAPER

About this paper

- This is a **practice past paper**, written in the style of NCEA excellence questions. It is **not** an official NZQA paper, and it may or may not be similar to the actual exam.
- Questions aim at the excellence end of the standard, so expect them to be harder than an average exam question. Don't be discouraged if some take a while.
- **Questions 21 onwards are a bit harder** than the earlier questions in this paper.

How to use it

- Try each question on your own first, ideally under timed, exam-like conditions.
- Show full working and give clear final answers, as you would in the real exam.
- Only check the answers once you've made a proper attempt.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

1. Solve the equation

$$3^{2x+1} - 28\,(3^x) + 9 = 0$$

You must use algebra to solve this problem.

2. Solve the equation

$$\left(x^2 - 3x\right)^2 - 2\left(x^2 - 3x\right) - 8 = 0$$

You must show algebraic working.

3. Solve the equation

$$2\log_4(x+2) - \log_2(x-4) = 2$$

Justify any solution(s) you reject.

4. If $\log_a x = 3$ and $\log_b x = 4$, find the exact value of $\log_{ab} x$.

Justify your answer, using correct mathematical statements.

5. Solve the equation

$$3^{x-1} = 5 \times 2^{x+1}$$

giving your answer exactly in the form $x = \dfrac{\log p}{\log q}$, where p and q are rational numbers.

6. The number of bacteria, N, in a sample t hours after it was prepared can be modelled by a function of the form $N = Ar^t$, where A and r are constants.

   After 3 hours there are 4000 bacteria in the sample. After 7 hours there are 25 000 bacteria in the sample.

   How many hours after the sample was prepared will the number of bacteria first exceed 100 000?

7. Aroha invests $5000 in an account that pays 4.2% interest per year, compounded annually.

At the same time, Ben invests $6500 in an account that pays 2.9% interest per year, compounded annually.

After how many complete years will Aroha's investment first be worth more than Ben's investment?

You must use algebra to solve this problem.

8. The intensity of light, I, at a depth of d metres below the surface of a lake can be modelled by a function of the form

$$I = I_0 \, k^d$$

where I_0 is the intensity of light at the surface, and k is a constant.

At a depth of 3 metres, the intensity of light is 60% of the intensity at the surface.

A diver's camera needs at least 10% of the surface intensity to take a clear photograph. What is the greatest depth at which the camera can take a clear photograph?

9. Show that the equation

$$kx^2 + (k+3)x + 1 = 0$$

where k is a non-zero real constant, has two distinct real roots for all values of k.

10. Use algebra to show that the equation

$$(x-a)(x-b) = k^2$$

where a, b, and k are real numbers, will always have real solutions.

Under what condition(s) will the equation have exactly one solution?

11. The line $y = mx - 1$ is a tangent to the curve $y = x^2 + 3x + 3$.

Find the possible values of m, and the coordinates of each point where the line touches the curve.

12. The equation

$$mx^2 + 4x + (m - 3) = 0$$

has two distinct real roots, both of which are negative.

Find the range of possible values of m.

13. The two roots of the equation $x^2 + bx + 10 = 0$ differ by 3.

Find the possible values of b.

14. The roots of the quadratic equation $ax^2 + bx + c = 0$ are in the ratio $2 : 3$.

Show that $6b^2 = 25ac$.

15. Find all value(s) of k for which the equation

$$\frac{1}{x-1}+\frac{1}{x+1}=k$$

has exactly one solution.

Justify your answer fully, using correct mathematical statements.

16.

$$\frac{x^2 + kx - 12}{x^2 - 16} = \frac{x - 3}{x - 4}$$

where k is a constant.

Find the value of k, and state any value(s) of x for which the equation is not valid.

17. A group of friends rent a bach for $360, and share the cost equally.

At the last minute, four more friends join the group. As a result, the cost to each person decreases by $15.

How many friends were in the original group?

You must use algebra to solve this problem.

18. A boat travels 24 km upstream against a current, and then 24 km back downstream with the current. The total time for the trip is 5 hours.

The current flows at a constant speed of 2 km h^{-1}.

Find the speed of the boat in still water.

You must use algebra to solve this problem. (Hint: speed $= \dfrac{\text{distance}}{\text{time}}$)

19. A parabolic arch is 30 m wide at ground level, and has a maximum height of 12 m.

A rectangular sign, 2 m high, is to be hung inside the arch so that its two top corners touch the arch, and its bottom edge is 7 m above the ground.

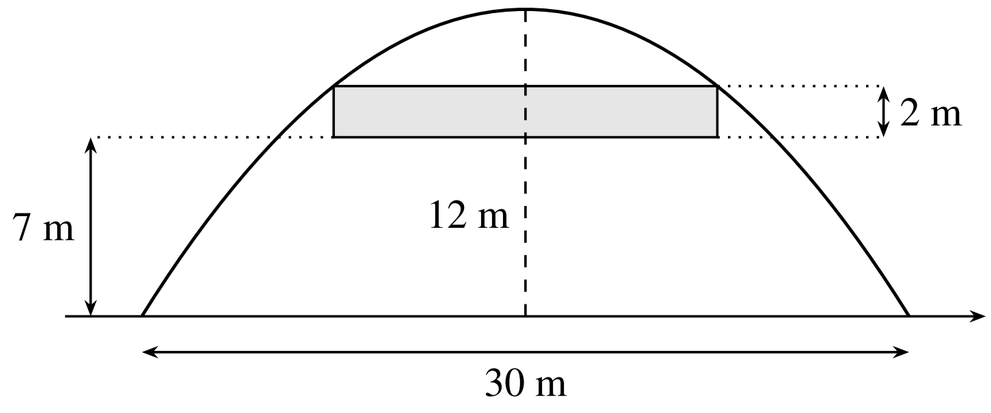


Diagram is NOT to scale

Find the width of the sign.

20. A rectangular garden measures 12 m by 8 m. A path is built around the garden.

The path is x m wide along the two longer sides of the garden, and $2x$ m wide along the two shorter sides.

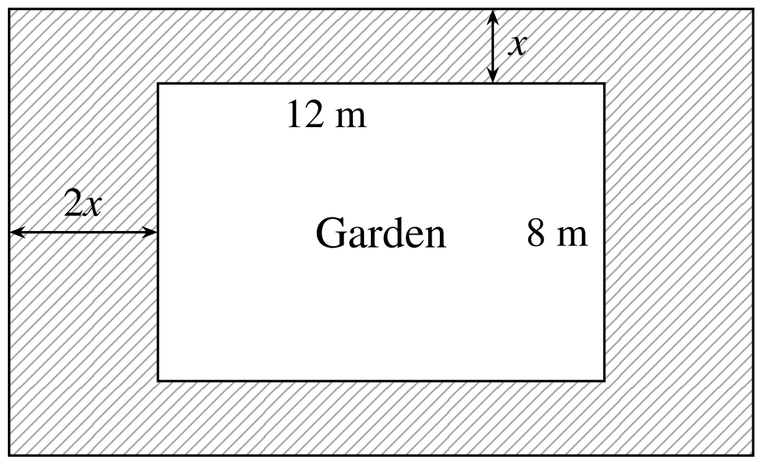


Diagram is NOT to scale

The area of the path is equal to the area of the garden.

Find the value of x, correct to 2 decimal places.

21. Solve the following pair of simultaneous equations:

$$2^x \times 4^y = 64$$

$$\frac{3^x}{9^y} = \frac{1}{27}$$

You must use algebra to solve this problem.

22. Solve the following pair of simultaneous equations, where $x > 0$ and $y > 0$:

$$\log_2 x + \log_4 y = 4$$

$$xy = 32$$

23. Find the range of values of x for which

$$(x-1)(x+3) > 2x+3$$

Give your answer in exact form.

24. Solve the equation

$$x^{\frac{2}{3}} - 5x^{\frac{1}{3}} + 6 = 0$$

You must use algebra to solve this problem.

25. Solve the equation

$$4^x + 4^{1-x} = 5$$

You must use algebra to solve this problem.

26. Solve the equation

$$\log_x 4 + \log_2 x = 3$$

(Hint: $\log_x 4 = \dfrac{\log_2 4}{\log_2 x}$)

27. Prove that the product of any four consecutive integers, plus one, is always a perfect square.

(Hint: let the four consecutive integers be n, $n + 1$, $n + 2$, and $n + 3$.)

28. Given that

$$x + \frac{1}{x} = 5$$

find the exact values of

(i) $x^2 + \frac{1}{x^2}$

(ii) $x^3 + \frac{1}{x^3}$

without finding the value of x.

29. The formula

$$y = \frac{2x + 3}{x - 1}$$

is defined for all values of x except $x = 1$.

Make x the subject of the formula, and hence explain why there is one value that y can never take.

30. The equations

$$x^2 + kx - 6 = 0 \qquad \text{and} \qquad x^2 - 5x + 6 = 0$$

have exactly one root in common.

Find the possible values of k.

31. Find the value(s) of k for which the equation

$$x^4 - (k+1)x^2 + k = 0$$

has exactly three distinct real solutions. Justify your answer.

32. The magnitude, M, of an earthquake on the Richter scale is given by

$$M = \log_{10}\left(\frac{A}{A_0}\right)$$

where A is the amplitude of the seismic waves, and A_0 is a fixed reference amplitude.

An earthquake in Canterbury had a magnitude of 7.2. An aftershock had a magnitude of 5.8.

How many times larger was the amplitude of the main earthquake than the amplitude of the aftershock?

33. Carbon-14 decays exponentially, so that the amount remaining halves every 5730 years.

A bone found at an archaeological site contains 18% of the carbon-14 it contained when the animal died.

How long ago did the animal die?

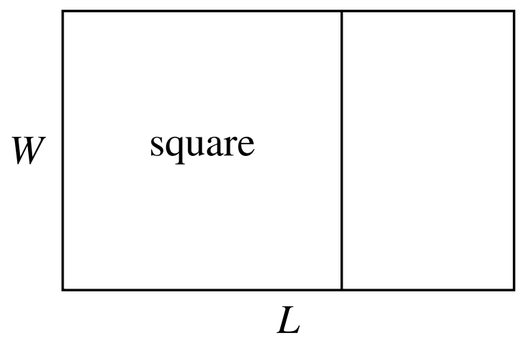
34. A right-angled triangle has a hypotenuse of length 13 cm, and a perimeter of 30 cm.

Find the lengths of the other two sides.

You must use algebra to solve this problem.

35. A rectangle has length L and width W, where $L > W$.

When a square with sides of length W is cut off one end of the rectangle, the rectangle that is left has exactly the same shape (i.e. is in the same ratio) as the original rectangle.



Find the exact value of the ratio $\frac{L}{W}$.

36. A theatre finds that when the price of a ticket is $\$p$, the number of tickets sold for a show is $600 - 20p$.

The theatre needs to receive at least \$4000 from ticket sales for each show.

Find the range of ticket prices that will allow this.

You must use algebra to solve this problem.

37. Ella claims that there is a value of k for which the roots of the equation

$$x^2 - kx + 2k = 0$$

are real numbers, and are reciprocals of each other (i.e. the roots are α and $\frac{1}{\alpha}$).

Is Ella correct? Justify your answer.

38. Solve the equation

$$2^{x^2} = 8 \times 4^x$$

You must use algebra to solve this problem.

39. Solve the equation

$$x^{\log x} = 100x$$

where $\log x$ means $\log_{10} x$, and $x > 0$.

40. The minimum value of the expression

$$x^2 - 2kx + 3k$$

is 2.

By completing the square, find the possible values of k.